

Mindfulness Anywhere: Meditating in the Generated Virtual World

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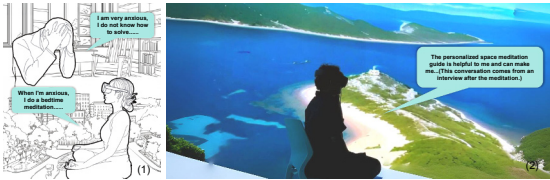


Figure 1: Generating a meditation environment in VR.

ABSTRACT

With the rapid advancement of Virtual Reality (VR) technology, immersive meditation experiences have become a promising tool for enhancing mental well-being. However, many existing VR meditation environments lack personalization and realistic integration, limiting their effectiveness. This study evaluated personalized virtual relaxation environments using a comparative research design. Results showed that the experimental group, using generative virtual meditation spaces, experienced significantly more positive emotional states. Future work will explore the impact of VR meditation through fNIRS measurements of prefrontal cortex activity, aiming to understand how personalized virtual environments influence psychological and neural responses.

Index Terms: VR Content, Meditation, Mental Health, Immersive Environments

1 INTRODUCTION

Meditation is widely recognized to improve mental health, reduce stress and anxiety, improve emotion regulation, and promote general well-being. Long-term practice induces positive changes in brain structure and function, supporting cognitive and emotional stability. Similarly, natural environments contribute significantly to mental health [5]. However, few studies integrate mindfulness with restorative experiences such as nature exposure [3]. Most VR meditation systems use static environments and predefined content, limiting adaptability, engagement, and long-term effectiveness. To address this, we develop personalized VR meditation experiences tailored to user preferences, creating dynamic environments that align with psychological needs and foster deeper self-exploration and healing.

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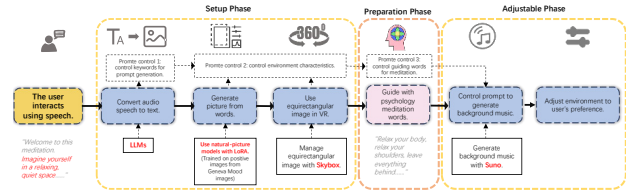


Figure 2: This system aims to provide customized, immersive meditation experiences for users.

2 METHODS

Content generation is fundamental to virtual reality, providing the foundation for immersive experiences. These experiences are critical in eliciting emotional responses by reflecting users' emotional expressions and distinguishing between positive and negative emotions. In mindfulness and meditation, the realism created by interactive VR environments, such as calming landscapes, enhances the user's sense of presence, helping them focus, relax and engage more deeply in their practice. This realism bridges the virtual and real, increasing the impact of meditation sessions [8].

2.1 Interactions Based on User Preferences

User-generated content (UGC) improves personalised experiences by prioritising user preferences [2]. Inspired by Carl Rogers' "Client-centred Therapy," which emphasises user-centric design [9], our system integrates user choices as a core element [1]. This approach enhances accessibility, engagement, and immersion through:

- **Speech-to-text and voice-driven customisation:** The system interprets user input to dynamically generate customised VR environments and meditation spaces customised to their preferences (see the Figure 1).
- **Generative environments:** create personalized virtual reality spaces and corresponding meditation music based on user input, enhancing the overall relaxation experience (see the Figure 2).

2.2 Immersion Based on Meditation

The system leverages the Blockad Skybox [6] to create customisable 360-degree VR environments guided by user input [10]. Speech-to-text technology ensures a seamless translation of user preferences into visual elements, enhancing the personalization of meditation spaces. Furthermore, immersive audio generated with Suno adapts to scenes and user interactions, creating realistic and contextually appropriate soundscapes [7]. Guided meditation scripts based on mindfulness meditation theory further support relaxation and well-being, integrating visual, auditory, and psychological elements for a deeply immersive and user-driven experience.



Figure 3: The VR-360-degree scenes generated based on users' voice input.

3 USER STUDY

The study investigates the neurophysiological differences between generative and standard VR meditation systems using a design between subjects with multiple measurements. Participants first complete the PANAS questionnaire to assess their baseline emotional state, followed by three phases: a 5 minute resting baseline, a 15-20 minute meditation session, and a 5 minute recovery after the removal of the VR headset.

Participants are randomly assigned to the experimental group, experiencing dynamically generated VR environments, or the control group, using a fixed VR scene, with identical breathing patterns [4]. After the session, the participants completed the PANAS questionnaire to assess emotional changes (5-point scale) and the UEQ to assess their VR experience (7-point scale).

4 KEY FINDINGS AND SUMMARY

A total of 16 participants (8 men, 8 women) participated, with ages ranging from 16 to 47 years (mean age 25.94, $SD = 6.49$). The standard group had a mean age of 27.88 years ($SD = 7.98$), while the experimental group had a mean age of 24 years ($SD = 3.61$).

Positive and Negative Affect Schedule (PANAS). The experimental group showed a significant increase in Positive Emotion scores ($M = 3.44, SD = 0.30, t = -4.134, p < 0.001$) and a significant decrease in Negative Emotion scores ($M = 1.42, SD = 0.31, t = 4.920, p < 0.001$). In contrast, the standard group did not show significant changes in Positive Emotions ($M = 2.31, SD = 0.52$) or Negative Emotions ($M = 1.50, SD = 0.48$).

These findings suggest that the generative VR meditation system effectively enhanced positive emotions and reduced negative emotions, likely due to its dynamic and personalised features. In comparison, the static design of the standard system may have limited its emotional impact. Figure 4 illustrates these differences, highlighting the critical role of system interactivity in influencing emotional outcomes. More research is needed to explore how specific features of the system drive these effects.

User Experience Questionnaire (UEQ). The experimental group scored significantly higher in Attractiveness ($M = 5.57, SD = 0.79, t = -6.433, p < 0.001$), Efficiency ($M = 5.50, SD = 0.79, t = -5.066, p < 0.001$), and Novelty ($M = 5.31, SD = 0.82, t = -4.581, p < 0.001$) compared to the standard group. These results indicate that the generative VR system provided a more engaging and innovative user experience, particularly in terms of visual appeal, usability, and novelty.

No significant differences were found between the groups in Perceptibility ($M = 4.21, SD = 0.49, p = 0.217$), Dependability ($M = 3.75, SD = 0.52, p = 0.078$), or Stimulation ($M = 3.35, SD = 0.73, p = 0.371$), suggesting that both systems performed similarly in clarity, reliability, and excitement.

Figure 5 illustrates these findings, highlighting the superior scores of the experimental group in engagement-related aspects and the potential of generative VR systems to enhance user experiences

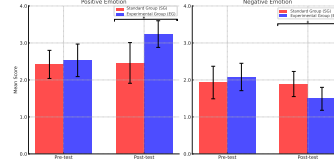


Figure 4: The PANAS scale scores for participants.

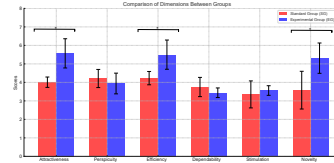


Figure 5: The UEQ scale scores for participants.

through dynamic and innovative features while maintaining comparable performance in other core usability dimensions.

Future work will focus on using fNIRS to investigate the neural mechanisms underlying the effectiveness of generative versus standard VR meditation environments. Specifically, the study will measure prefrontal cortex activity during structured box breathing, with a focus on oxygenated hemoglobin (HbO) and deoxygenated hemoglobin (HbR) levels to assess cognitive control, emotional regulation, and stress reduction. By comparing neural activation patterns between the two groups, future research aims to identify how generative VR environments support well-being.

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